Chimdiume Azie

AI/ML Developer · B.Sc. Computer Science

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Profile

Recent Computer Science graduate (Godfrey Okoye University, 2026) focused on building explainable AI systems end to end. Built a deep learning system that classifies Alzheimer's stages from brain MRI scans and explains its predictions, together with the hospital web application around it, designed to run on ordinary CPU hardware. Also building Nellow, an AI agent with a custom 3D web interface. Seeking an entry-level AI/ML or software engineering role.

Projects

Alzheimer's Disease Detection System using Deep Learning and Explainable AI (Final-year project, github.com/Achimdi/neuroscan-ai)

- Trained an EfficientNet-B0 classifier (PyTorch, transfer learning) to sort brain MRI scans into four stages (NonDemented, VeryMildDemented, MildDemented, ModerateDemented) using the OASIS MRI dataset.
- Added Grad-CAM heatmaps that show which brain regions drive each prediction; for Mild and Moderate cases they concentrated on the hippocampus, medial temporal lobe and ventricles.
- Built a Flask hospital web application with Admin and Doctor roles (JWT authentication, role-based access, one-time registration codes), patient records, MRI upload with predicted stage, confidence score and heatmap, saved result history, and an audit trail for patient transfers (SQLite, SQLAlchemy).
- Handled class imbalance with a class-weighted loss, and trained and ran the whole system on a CPU-only laptop to suit resource-limited hospitals.
- Results: 98.66% validation accuracy. Accuracy on the held-out test set was 77.35%; I traced the gap to a skewed original split (88.5% NonDemented in the test set against 40.8% in training) and documented stratified splitting and clinical validation as next steps. Research prototype, not clinically validated.

Nellow: AI Agent (github.com/Achimdi/nellow)

- Designed and built an AI agent that runs locally, with a web interface featuring 3D animations and motion effects.
- Currently redesigning the interface as a futuristic, metallic design system, with hosting planned.

Skills

- **Machine learning and AI:** PyTorch, transfer learning, CNNs, Grad-CAM and explainable AI, scikit-learn, OpenCV, NumPy, Pandas, model evaluation (precision, recall, F1, confusion matrix, ROC)
- **Web and backend:** HTML, CSS, JavaScript, Flask, REST APIs, JWT authentication, SQLite and SQLAlchemy
- **Mobile:** Flutter (working knowledge)
- **Tools:** Python, Git and GitHub, VS Code

Education

Bachelor of Science (B.Sc.) in Computer Science, Godfrey Okoye University, 2026

Relevant coursework: Expert Systems, Computer Networks and Communication, Compiler Construction, Structured Programming, Human Computer Interaction, Systems Analysis and Design

References

Available on request.